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Para la falla A calcule.

x_1, y_1
(60, 50)

Falla A.

x_2, y_2
(100, 40)

Datos:

x_2, x_1

100 - 60

y_2, y_1

40 - 50

Sitio (75, 20)

x_2, x_1

$x - 60$

y_2, y_1

$y - 50$

Paso 1. Longitud de la falla

$$L_A = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$L_A = \sqrt{(100 - 60)^2 + (40 - 50)^2}$$

$$L_A = 41.23 \text{ km}$$

Paso 2. Ecuación de la recta

$$(y_2 - y_1) = m(x_2 - x_1)$$

$$m = \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

$$(y - 50) = m(x - 60)$$

$$m = \left(\frac{40 - 50}{100 - 60} \right)$$

$$(y-50) = \left(\frac{40-50}{100-60} \right) (x-60)$$

$$y-50 = -\frac{1}{4} (x-60)$$

$$y-50 = -\frac{x+60}{4}$$

$$4y-200 = -x+60$$

$$4y-200+x-60=0$$

$$x+4y-260=0$$

Paso 3 Distancia al sitio de la falla

$$|R| = \left| \frac{Ax + By + C}{\sqrt{A^2 + B^2}} \right|$$

$$|R| = \left| \frac{x + 4y - 260}{-\sqrt{1^2 + (4)^2}} \right|$$

$$|R| = \left| \frac{x + 4y - 260}{-\sqrt{17}} \right|$$

• Sitio $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 75 \\ 20 \end{pmatrix}$

$$|R| = \left| \frac{75 + 4(20) - 260}{-\sqrt{17}} \right|$$

$$|R| = | 25.4 \text{ (dkm)} |$$

Paso 4: Magnitud momento M_w

1. Transcurrente (strike-slip)

$$M_w = 5.16 + 1.12 \log(L)$$

$$M_w = 5.16 + 1.12 \log(41.23)$$

$$M_w = 6.969$$

2. Falla inversa

$$M_w = 5 + 1.22 \log(L)$$

$$M_w = 5 + 1.22 \log(41.23)$$

$$M_w = 6.971$$

3. Falla Normal

$$M_w = 4.86 + 1.32 \log(L)$$

$$M_w = 4.86 + 1.32 \log(41.23)$$

$$M_w = 6.992$$

Paso 5: Falla transcurrente.

Datos:

$$M = 6.969$$

$$R = 25.466$$

Joyner y Boore (1993) $PGA = 0.090g$

$$\log(PGA) = -1.229 + 0.227 M - \log \sqrt{R^2 + 44.225} - 0.0023 \times \sqrt{R^2 + 44.225}$$

$$\log(PGA) = -1.229 + 0.227 (6.969) - \log \sqrt{(25.466)^2 + 44.225} - 0.0023 \times \sqrt{(25.466)^2 + 44.225}$$

Climont et al. (1994)

$$\ln(PGA) = -1.687 + 0.593 \mu - 0.37 \ln(R) - 0.00202 R$$

conversione m/s² og

$$PGA = 0.154g$$

Corbell et al. (1979)

$$\ln(PGA) = 6.74 + 0.859(6.969) - 1.8 \ln(25.469 + 25)$$

conversione m/s² og

$$PGA = 0.334g$$

Campbell (1981)

$$\ln(PGA) = -4.41 + 0.868(6.969) - 1.09 \ln(25.469 + 0.609 e^{0.74})$$

$$PGA = 0.163g$$

Problema PGA Falla Inversa.

Datos: $\mu = 6.971$

$$R = 25.466$$

Jagner Boore (1993)

$$\log(PGA) = -1.29 + 0.727(6.971) - \log \sqrt{(25.466)^2 + 44.225} - 0.0023 \sqrt{(25.466)^2 + 44.225}$$

$$PGA = 0.090g$$

Climent et al (1994)

$$\ln(PGA) = -1.687 + 0.553(6.969) - 0.537 \ln(R) - 0.00302R$$

conversion de m/s^2 a g

$$PGA = 0.155g$$

Cornell et al (1979)

$$\ln(PGA) = 6.74 + 0.859(6.969) - 1.8 \ln(25.466 + 25)$$

conversion de m/s^2 a g

$$PGA = 0.335g$$

Compbell (1981)

$$\ln(PGA) = 4.141 + 0.868(6.969) - 1.09 \ln(25.466 + 0.0606e^{0.7M})$$

$$PGA = 0.164g$$

Paso 7 PGA Falla Normal

Datos: $M = 6.992$

$$R = 25.466$$

Jayner Moore (1993)

$$\log(PGA) = 1.229 + 0.227(6.992) - \log \sqrt{(25.466)^2 + 44.225}$$

$$PGA = 0.091g$$

Climent et al. (1994)

$$\ln(PGA) = -1.687 + 0.553(6.992) - 0.537 \ln(25.466) - 0.00302(25.466)$$

Conversion cm/s^2 a g .

$$PGA = 0.156g$$

Cornell et al. (1979)

$$\ln(PGA) = 6.74 + 0.854(6.992) - 1.8 \ln(25.466) + 25)$$

Conversion cm/s^2 a g

$$PGA = 0.340g$$

Campbell (1981)

$$\ln(PGA) = -4.141 + 0.868(6.992) - 1.04 \ln(25.466 + 0.0606 e^{0.74})$$

$$PGA = 0.167g$$

Tipo Falla	Distancia a Epicentro (km)	Mw	PGA			
			Joyner Boore 1993	Clement et al 1994	Cornell et al 1979	Campbell 1981
Transversante Strike Slip	25.4166	6.869	0.040g	0.154g	0.334g	0.163g
Inversa	25.4166	6.971	0.040g	0.155g	0.335g	0.164g
Normal	25.4166	6.992	0.091g	0.156g	0.340g	0.167g